Bandwidth Estimation: Ingress, Egress, Gbps & CDN Offload

data per second, and why video lives on CDNs

Watch the 3-min video

In one line: bandwidth = requests per second (or concurrent streams) × payload size, converted from bytes to bits, then sized for the peak and reduced by CDN offload; 10,000 streams at 5 Mbps is 50 Gbps of egress.

1 Ingress and egress

Your video platform has **10,000 people watching at the same time**, each stream at **5 megabits per second**. Can one data-center link handle that? What will the monthly bill look like? Bandwidth is often the most expensive line in a video platform's budget, so you need to estimate it.

Bandwidth is data per second. Ingress is data coming into your system: uploads and API requests. **Egress** is data going out: page loads, downloads and video. For most consumer apps egress is much larger, and cloud providers usually charge for egress, not ingress.

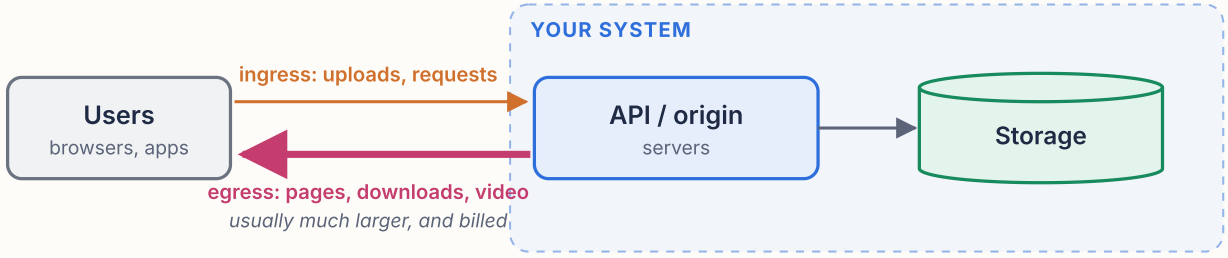
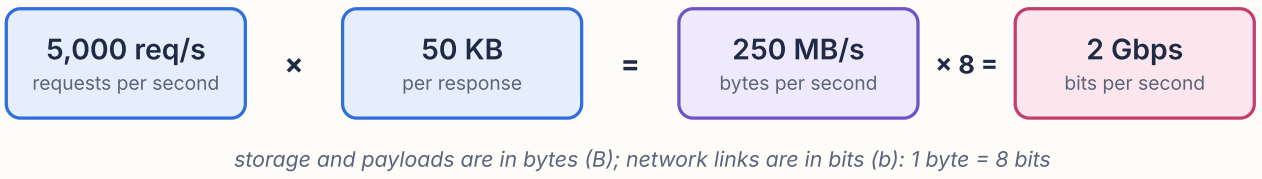


Figure 1. Requests are small and come in; responses are large and go out. Egress is the number that drives cost.

2 The formula, and the bits-vs-bytes trap

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bandwidth = requests per second × payload size  ·  Gbps = GB/s × 8
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For example, 5,000 requests per second × 50 KB per response = **250 MB/s**. Network links are measured in **bits**, so multiply bytes by 8: 250 MB/s × 8 = 2,000 Mbps = **2 Gbps**. For streaming, use concurrent streams × bitrate instead; bitrates are already in bits.



storage and payloads are in bytes (B); network links are in bits (b): 1 byte = 8 bits

Figure 2. Bytes per second from the app; bits per second for the network. Forgetting the × 8 gives an answer 8 times too small.

3 Interview question: 10,000 concurrent streams

WORKED EXAMPLE

Rate: 10,000 streams × 5 Mbps = 50,000 Mbps = **50 Gbps** of egress.

Hardware: a typical server network card is 10 or 25 Gbps. Even at full line rate, 50 Gbps needs 2 × 25 Gbps or 5 × 10 Gbps cards; with headroom, several servers just for the network.

Monthly volume: 50 Gbps ÷ 8 = 6.25 GB/s. A 30-day month has about 2.6 million seconds, so 6.25 GB/s × 2.6M s ≈ **16 PB** of traffic a month, "over 15 petabytes". Even at an illustrative 1 cent per GB, that is about $160,000 a month for this one stream of traffic.

4 CDN offload

That is why video platforms are dominated by **CDN egress**. The CDN stores popular videos at edge locations near viewers and serves most bytes from there. With a **95% offload**, the origin serves only 5%: **2.5 Gbps instead of 50**. The CDN bill is still large, but much cheaper per GB, and viewers get lower latency because the bytes travel a shorter distance.

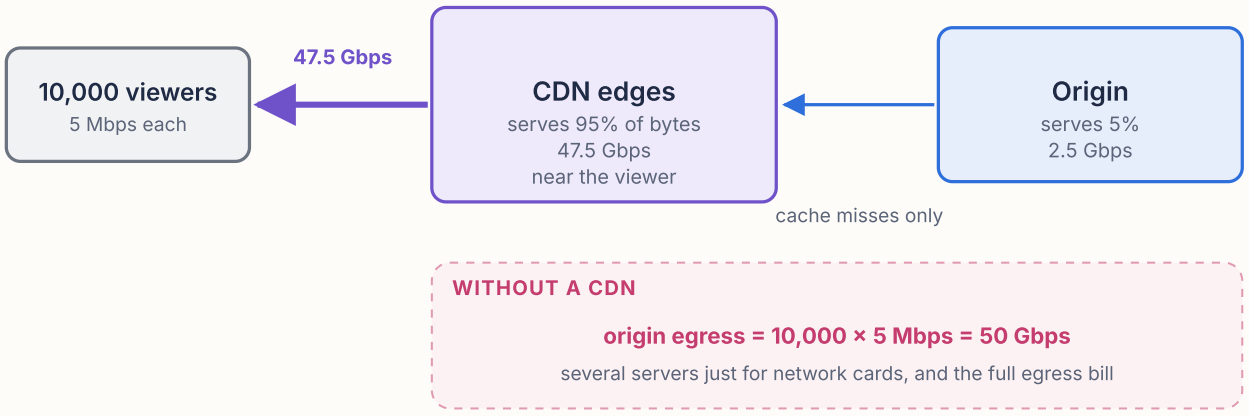


Figure 3. With 95% offload, the origin's egress drops from 50 Gbps to 2.5 Gbps; the CDN carries the rest from edges close to viewers.

5 Peaks and cross-region transfer

- **Size for the peak, not the average.** Evening prime time can be several times the daily average, and a big live event many times more. If prime time is 3× the average, 50 Gbps average means links and CDN commitments for about 150 Gbps.
- **Cross-region transfer is billed.** Copying data between regions or availability zones costs money. Replicating every upload to three regions multiplies that cost.

6 Ways to cut bandwidth

- **Compress responses** with gzip or brotli; JSON and HTML often shrink several times.
- **Modern codecs** for images and video (WebP or AVIF images; H.265, VP9 or AV1 video) send fewer bits for the same quality.
- **Adaptive bitrate** streaming matches the bitrate to the viewer's screen and network: no 4K stream to a phone on 4G.
- **Cache aggressively at the edge.**
- **Paginate** API responses, and **don't send fields** the client never uses.

COMMON MISTAKES

- Forgetting to multiply bytes by 8 when comparing with a network link.
- Sizing for the average instead of prime time or a live event.
- Estimating ingress and ignoring egress, which is where the bill is.
- Forgetting cross-region and cross-zone transfer charges.
- Serving video from the origin instead of a CDN.

7 Key terms

Bandwidth: data per second (Mbps, Gbps)

Bitrate: bits per second of a stream

Peak factor: peak ÷ average traffic

Ingress: data into your system

CDN offload: share of bytes served by the CDN

NIC: network card, e.g. 10 or 25 Gbps

Egress: data out; usually billed

Origin: your own servers behind the CDN

Adaptive bitrate: stream quality follows the network

8 Interview questions

Q1. How do you estimate bandwidth? BASIC

Requests per second (or concurrent streams) × payload size, converting bytes to bits, then apply a peak factor and the CDN offload.

Q2. What is the difference between ingress and egress, and which costs more? BASIC

Ingress is data coming in (uploads, requests); egress is data going out (pages, video). Egress is usually much larger, and clouds usually bill egress, not ingress.

Q3. 5,000 RPS with 50 KB responses. What bandwidth? SCENARIO

5,000 × 50 KB = 250 MB/s × 8 = 2 Gbps. That fits one 10 Gbps card, but plan for peak and use several servers for redundancy.

Q4. 10,000 concurrent streams at 5 Mbps. What egress, and what do you do about it? SCENARIO

50 Gbps, about 16 PB a month. Push it through a CDN: at 95% offload the origin serves only 2.5 Gbps.

Q5. Average egress is 20 Gbps and prime time is 4× average. What do you provision? SCENARIO

Size links and CDN commitments for about 80 Gbps, plus headroom for a live event, not 20 Gbps.

Q6. Your API responses are 2 MB each and bandwidth is high. What can you change besides adding links? TRICKY

Compress with gzip or brotli, paginate, remove unused fields, cache at the edge. A 2 MB JSON response often shrinks to a few hundred KB compressed.

Q7. You replicate every upload to three regions. What happens to cost? TRICKY

Each upload is transferred to two other regions, and cross-region transfer is billed, so that transfer cost is paid twice per upload, on top of 3× the storage. Replicate only what needs it.

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